concentration gradients tend to disappear, the large temperature differences in operating nuclear fuel can result in a lower chemical potential at a higher concentration leading to an increase in the concentration gradient. Hence, some fission products, e.g. the noble gases, Cs and I, migrate to the fuel – cladding gap (lowest temperature), whereas others, e.g. Zr and Nb, migrate to the center line (highest temperature), cf. Fig. 21.5(b). A penetration of the can during reactor operation thus leads to an initial rapid release of those fission products which have accumulated in the gap followed by a slow release of fission products present in the cracks and at grain boundaries and finally by a much slower dissolution and release of fuel (U, other actinides and other fission products).

The high concentration of uranium (and/or Pu) atoms in the fuel in combination with strong resonance peaks at certain neutron energies also leads to self-screening effects (n-flux depression at energies with large reaction cross sections), especially for $^{238}U(n,\gamma)$, $^{239}Pu(n,\gamma)$ and $^{239}Pu(n,f)$ (see Fig. 19.3). Thus, most of the Pu formed is located near the fuel surface and relatively little at the fuel center. This is easily seen in α-autoradiographs of spent fuel pins where most of the α:s are found in a ring near the fuel surface. Hence, thinner fuel pins yield a better fuel utilization than thicker pins, but thinner pins means higher fuel fabrication costs for the same amount of uranium. In practice an economic compromise usually results in pin diameters slightly less than 1 cm.

Radiation effects and oxidation causes changes in the tensile properties of the canning material. Fresh fuel has a can that is very ductile whereas the can of spent high burn-up fuel normally is hard and brittle.

In an operating power reactor, only part of the fuel is replaced annually; e.g. 1/6 to 1/4 of the total number of fuel elements. The most burnt up fuel elements are removed from the core as spent fuel and replaced by fresh fuel. In order to achieve as even as possible heat generation in the core (permitting the highest power output), the fresh fuel elements are mostly loaded in the outer core regions whereas partly spent fuel is moved in towards the center. This results in a checker board pattern of fuel of varying age in the reactor. However, several variants to these movements are in use.

21.3. COMPOSITION AND PROPERTIES OF SPENT FUEL ELEMENTS

The composition of spent reactor fuels varies as a function of input composition (kinds and amounts of fissile and fertile atoms), neutron spectrum, flux, fluency (or burnup), design of the pins and fuel elements, positions occupied in the reactor during operation, and the cooling time after removal from the reactor. A harder neutron spectrum increases fertile to fissile conversion (Fig. 19.3). Hence, after refuelling, some BWRs are initially operated at the highest possible void fraction in order to maximize conversion of ^{238}U to plutonium. This permits a higher final burnup of the fuel. Increased burnup increases the concentration of fission products and larger amounts of higher actinides are formed (Fig. 19.7). Thinner fuel pins increases conversion to and burning of plutonium due to

less self screening. A high neutron flux results in more high order reactions (§17.3), while a long irradiation time produces relatively larger amounts of longlived products. With increased cooling time the fraction of shortlived products is reduced.

Because of such effects, spent uranium fuel elements from PWR, BWR, HWR, GCR and FBR differ in composition both from each other and between fuel batches from the same reactor. Furthermore, the composition differs between pins in the same fuel element and for each pin also along its length, especially when initial burnable poison concentration and enrichment is graded along pins. The difference is not so large that very different fuel cycles (e.g. other reprocessing schemes) are required as long as the fuel is based on uranium metal or uranium dioxide. In the following subsections we mainly discuss uranium dioxide fuel elements, and, more specifically, LWR elements. Fission product and actinide yields for a typical BWR fuel are given in Table 21.2.[1] The uranium once through part of Figure 21.1 shows the annual materials flow in a mixed BWR–PWR conglomerate with a total average power of 10 GW_e (i.e. about twelve 1000 MW_e plants) running at full power for 7000 hours per year (load factor ~80%).

Table 21.2 Calculated average composition after 1 y cooling of 1 t U as 3.6% enriched UO_2 fuel with 60 MWd/kg U burnup in a typical BWR where the n-flux changes yearly after fuel rearrangement. Straight figures are weight in kg/t IHM and italics are radioactivity in TBq/t IHM. Total weight is 63 kg FPs and 14 kg transuranium elements

H																	He
Li	Be											B	C 0.14 *0.04*	N	O 134.5 *0*	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge $8\cdot10^{-4}$ *0*	As $2\cdot10^{-4}$ *0*	Se 0.096 *0.025*	Br 0.036 *0*	Kr 0.68 *1930*
Rb 0.44 $2\cdot10^{-6}$	Sr 1.3 *3900*	Y 0.72 *3900*	Zr 5.7 *54*	Nb $9\cdot10^{-5}$ *130*	Mo 6.2 *0*	Tc 1.1 *0.71*	Ru 4.8 *5900*	Rh 0.53 *5900*	Pd 3.6 *0.010*	Ag 0.12 *7.3*	Cd 0.30 $2\cdot10^{-5}$	In 0.001 $4\cdot10^{-13}$	Sn 0.12 *11*	Sb 0.028 *270*	Te 0.99 *0.004*	I 0.44 *0.002*	Xe 9.8 $4\cdot10^{-13}$
Cs 4.6 *15000*	Ba 3.3 $6\cdot10^{-8}$	*Ln	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl $3\cdot10^{-12}$ *0.041*	Pb $6\cdot10^{-7}$ $5\cdot10^{-4}$	Bi $3\cdot10^{-12}$ $5\cdot10^{-4}$	Po $2\cdot10^{-15}$ $8\cdot10^{-4}$	At $5\cdot10^{-22}$ $3\cdot10^{-8}$	Rn $1\cdot10^{-14}$ $5\cdot10^{-4}$
Fr $6\cdot10^{-18}$ $3\cdot10^{-8}$	Ra $2\cdot10^{-10}$ $5\cdot10^{-4}$	**An	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	113	Fl	115	Lv	117	118

*Lanthanides	La 2.2 $8\cdot10^{-7}$	Ce 4.3 *4000*	Pr 2.0 *4000*	Nd 7.2 $5\cdot10^{-10}$	Pm 0.061 *2100*	Sm 1.3 *9.3*	Eu 0.33 *1600*	Gd 0.52 *0.32*	Tb 0.006 *0.36*	Dy 0.003 $2\cdot10^{-13}$	Ho 0.006 *0*	Er	Tm	Yb	Lu
**Actinides	Ac $6\cdot10^{-11}$ $2\cdot10^{-7}$	Th $3\cdot10^{-6}$ *0.012*	Pa $7\cdot10^{-7}$ *0.042*	U 923 *0.19*	Np 1.2 *3.6*	Pu 12 *6500*	Am 0.74 *37*	Cm 0.44 *1500*	Bk $4\cdot10^{-7}$ *0.027*	Cf $2\cdot10^{-6}$ *0.005*	Es $8\cdot10^{-12}$ $6\cdot10^{-7}$	Fm $2\cdot10^{-19}$ $5\cdot10^{-12}$	Md	No	Lr

[1] As the exact composition of spent fuel varies considerably and depend on many factors, the reader will find slightly varying figures in this text.

21.3.1. Fission products

About 34 kg fission products (FP, including gaseous) are formed in each initial ton of uranium irradiated to 33 MWd/kg. To accommodate mixed oxide fuels (i.e. fuels initially containing both U and Pu), burnup, composition, activities, etc, are usually normalized to the amount of *initially present heavy metal*, *IHM*. The composition of spent fuel varies with burnup, power history and reactor. The formation rates of the various primary fission products depend on the fission rate, chain yields of the fissioning nuclide and on the corresponding charge distribution (Ch. 11). Increased fission of heavier actinides displaces the lower mass peak in the yield curve towards higher mass numbers while the heavier mass peak remains about the same (Fig. 11.9). An increased contribution from fast fission increases the yield in the valley between the peaks. Because of the continued n-irradiation, secondary n,γ-reactions occur with the primary fission products and their daughters (Ch. 17). As an example, ^{134}Cs is not formed to any appreciable extent in fission because it is shielded by the stable ^{134}Xe. Hence, no ^{134}Cs is normally observed in the remains after a nuclear explosion in the atmosphere. However, primary fission products in the $A = 133$ isobar chain have time to decay to stable ^{133}Cs during reactor operation and ^{134}Cs is produced by the reaction ^{133}Cs (n,γ) ^{134}Cs. Given the cooling time, the ratio between the decay rates of ^{134}Cs and ^{137}Cs can be used to estimate the burnup of fuel from a given reactor, see Figure 21.6.

Using effective cross-sections and yield values the amounts and radioactivities in Figure 21.7 and Table 21.2 were calculated. It is seen that Xe, Zr, Mo, Nd, Cs, and Ru, which are the elements formed in largest amounts in thermal fission (both by mole percent and by weight), constitute about 70% of the fission product weight after a cooling time of 1 y.

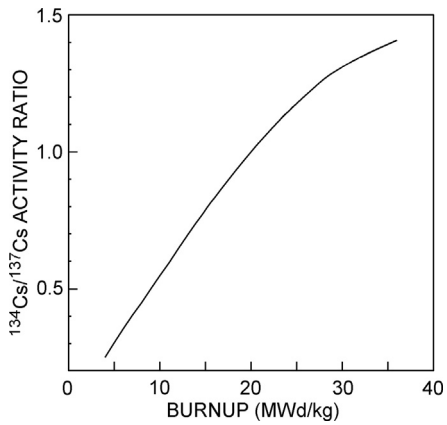


Figure 21.6 Typical variation of the ratio between the ^{134}Cs and ^{137}Cs radioactivities with fuel burnup.

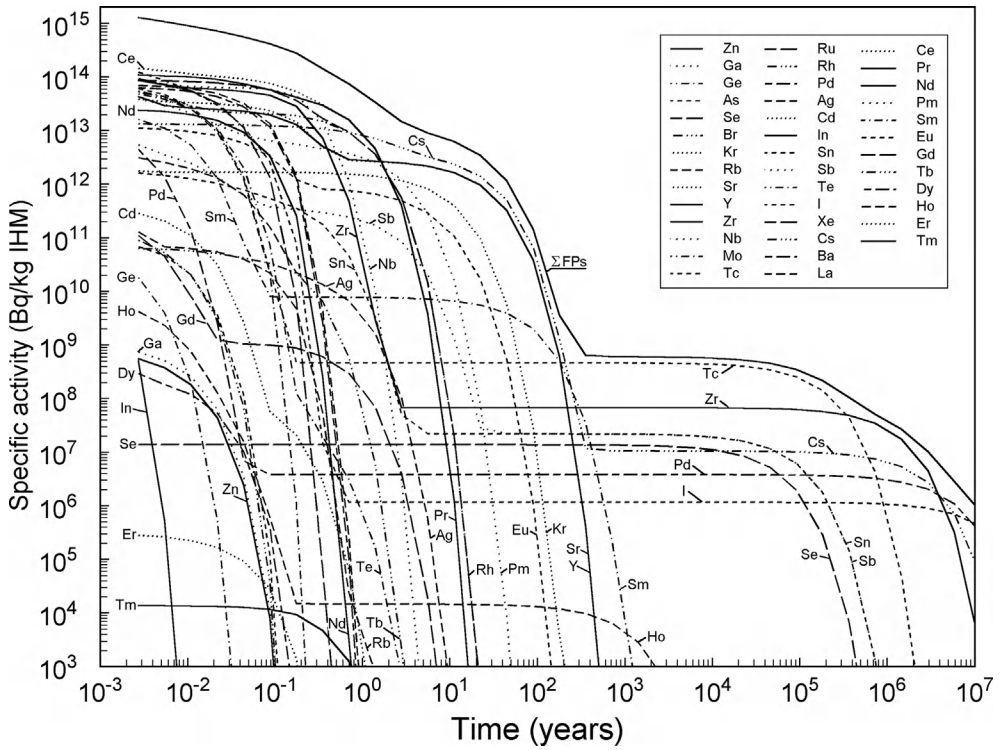


Figure 21.7 Radioactivity of fission products per kg IHM in spent PWR fuel at 33 MWd/kg burnup. Inflexion points indicate the existence of several radioisotopes.

At cooling times 10−1000 y the activities of ^{90}Sr and ^{137}Cs (with daughters) dominate among the fission products. Later the fission product activity is due to very longlived nuclides of low activity (Figure 21.7, Table 21.3). ^{131}I, which for very short cooling times is one of the most hazardous FPs because of its affinity to the thyroid gland, has practically disappeared after a cooling time of 6 months. The $A = 95$ isobar chain, which is formed in highest yield (6.52%), leads to ^{95}Zr ($t_{½}$ 64 d) → ^{95}Nb($t_{½}$ 35 d) → ^{95}Mo (stable).

From Table 21.2 it is seen that only stable isotopes remain for some fission elements at $t_{cool} \geq 1$ y, while some others are of low activity. Some of the more active ones after 1 y have disappeared almost completely at 100 y (T, Sb, Ce, Eu, Pm, Ru, Rh, and Kr), leaving essentially only the ^{90}Sr−^{90}Y and ^{137}Cs−^{137m}Ba pairs and ^{151}Sm.

21.3.2. Actinides

Neutron capture and β-decay lead to the formation of higher actinides. This is illustrated in Figs. 14.3, 14.4, 19.5 and 19.7. ^{239}Pu and ^{241}Pu also fission, contributing significantly

Table 21.3 Some long-lived radionuclides produced in fission; t_{cool} 1 y, data as in Table 21.02

Nuclide	Half-life (years)[†]	Decay; β^- (energy MeV)[‡]	Thermal fission yield(%)[††]	Activity (TBq/t U)
^{79}Se	$<6.5 \times 10^4$	0.1509	0.0443	251
^{85}Kr	10.72	0.6874	1.318	1930
^{87}Rb	4.8×10^{10}	0.2823	2.558	1.2×10^{-6}
^{90}Sr → ^{90}Y	28.5	0.5462 + 2.2815	5.772	3890
^{93}Zr → ^{93m}Nb	1.5×10^6	0.0905	6.375	0.112
^{99}Tc	2.13×10^5	0.2936	6.074	0.710
^{107}Pd	6.5×10^6	0.033	0.147	0.0104
^{126}Sn → ^{126m}2Sb | \ ↓ | ^{126m1}Sb | ↓ └→ ^{126}Sb	$\sim 1 \times 10^5$	0.368 + 3.670	0.0536	0.0493
^{129}I	1.57×10^7	0.192	0.757	0.0023
^{134}Cs	2.062	2.0585	0	8760
^{135}Cs	3.0×10^6	0.205	6.536	0.0261
^{137}Cs → ^{137}mBa	30.0	0.5134	6.183	6619
^{151}Sm	90	0.0763	0.4196	9.34
^{154}Eu	8.8	1.9689	0	984
^{155}Eu	4.96	0.2527	0.0320	574

[†]Only for the longer lived mother nuclide.
[‡]Decay energy, not particle energy (see decay schemes).
[††]Thermal fission of ^{235}U (fission of ^{238}U, fission of Pu isotopes and n,γ-reactions are important effects in a nuclear reactor).

to the energy production (Fig. 19.8). Truly, all plutonium isotopes lead to fission, since the n-capture products and daughters, ^{241}Am, ^{242}Am, ^{243}Cm, ^{245}Cm, and most other actinides are either fertile, fissible or fissile.

Fast neutrons in the reactor induce (n,2n) reactions, e.g. ^{238}U (n,2n) ^{237}U (β^-) ^{237}Np (n,γ) ^{238}Np (β^-) ^{238}Pu (n,γ) ^{239}Pu (see Fig. 19.5), as well as fast fission. ^{237}Np is also formed through the reaction ^{235}U (n,γ) ^{236}U (n,γ) ^{237}U (β^-) ^{237}Np. These reactions are the main sources of neptunium and of ^{238}Pu. Many actinides formed are fissile but have a short half-life. The ratio, x, of the amount fissioned to the amount decayed or reacted with neutrons at constant flux and steady state is given by

$$x = \varphi \sigma_f / \left[\lambda + \varphi\left(\sigma_f + \sigma_{n,\gamma}\right)\right] \tag{21.9}$$

where φ is the neutron flux, σ_f the effective fission cross-section, $\sigma_{n,\gamma}$ the effective cross-section for radiative capture and λ the decay constant. As can be seen from (21.9) a higher fraction of the nuclide will fission in a very high neutron flux ($x \to \sigma_f/(\sigma_f + \sigma_{n,\gamma})$ when $\varphi \to \infty$), whereas most of it will disappear by decay in a low flux ($x \to 0$ when $\varphi \to 0$). As an example, practically all ^{238}Np formed by ^{237}Np(n,γ) ^{238}Np will fission when

$\varphi\sigma_f \gg \lambda$ ($\sigma_f = 2070$ b, $\sigma_{n,\gamma} \sim 0$ b and $\lambda = 3.8 \times 10^{-6}\ s^{-1}$). Hence, the buildup of many higher actinides is less efficient in a high flux reactor than in a reactor with a more moderate neutron flux although the total buildup rate might still be higher in the higher flux. This is accentuated in fast breeder reactors where the combination of a very high neutron flux and a hard neutron spectrum (increased effective σ_f for fissible nuclides) strongly reduces the amount of higher actinides formed (at a given burnup) compared to a thermal reactor.

From the fission and capture cross-sections, and half-lives the radioactivity of each actinide element in one kg spent fuel (and radium) has been calculated and shown in Figure 21.8 for a PWR UO_2 fuel with a burnup of 33 MWd per kg IHM. Due to the use of a log-log scale in Figure 21.8, the decay curve of any single nuclides is s smooth curve bending downwards. Inflexion points indicate the existence of several radioactive isotopes of the same element with different half-lives.

The total amount of plutonium formed in various reactors is given in Table 21.4. The old gas-graphite reactors and heavy water reactors are the best thermal plutonium

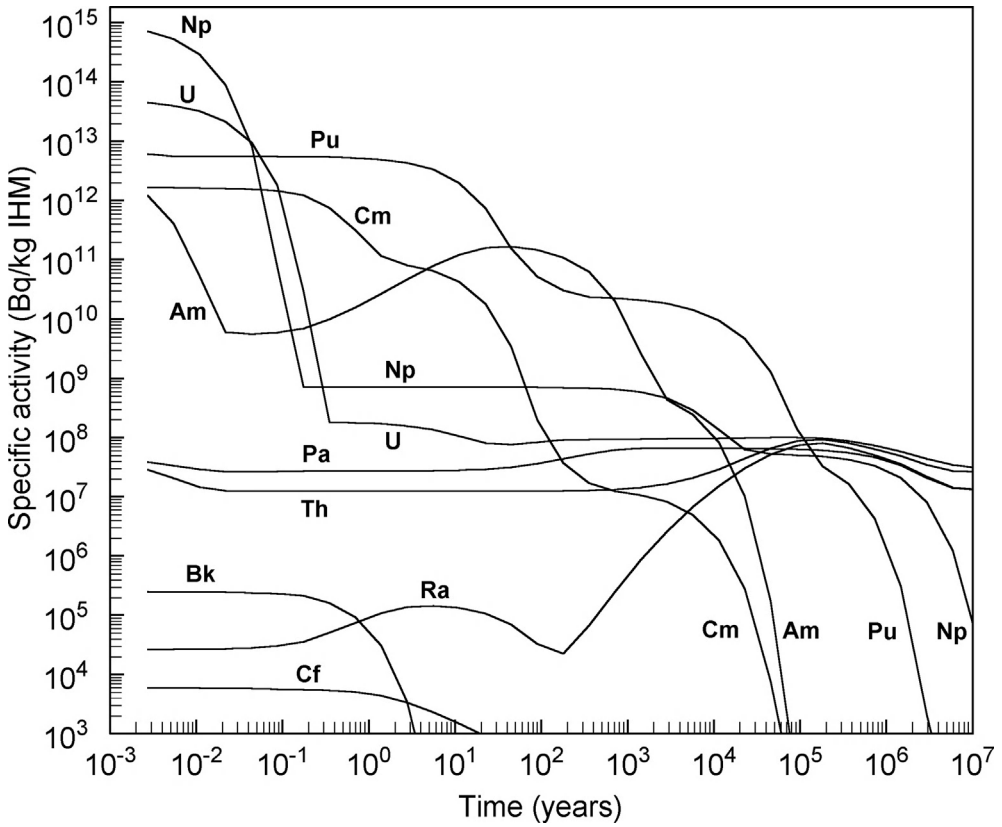


Figure 21.8 Radioactivities of actinides and radium per kg IHM in spent PWR fuel after 33 MWd/kg burnup. Inflection points indicate the presence of several radioisotopes of the element.

Table 21.4 Production of plutonium (kg/MW_ey) in various reactor types

Reactor type	Total Pu	Fissile Pu
Light water reactors	0.26	0.18
Heavy water reactors	0.51	0.25
Gas-graphite reactors	0.58	0.43
Advanced gas-cooled reactors	0.22	0.13
Liquid metal fast breeder reactors	1.35	0.7—1.0

producers. They have therefore been used in weapons fabrication. The fast breeder reactor is also an efficient Pu producer. Whereas thermal reactors (except at very low burnup) produce a mixture of odd and even A Pu isotopes, a fast breeder loaded with such a mixture, by a combination of fission and n-capture increases the relative concentration of Pu isotopes with odd A in the core and produces fairly pure ^{239}Pu in the blanket. Hence the combined Pu product from core and blanket elements has a much higher concentration of fissile Pu isotopes than plutonium from a thermal reactor (the fast breeder not only produces more Pu than it consumes but also improves Pu quality, i.e. increases the concentration of fissionable isotopes). The LWR and AGR are the poorest plutonium producers.

Composition of spent fuel varies somewhat with cooling time, e.g. the amount of ^{241}Am increases because it is the daughter of 14.4 y ^{241}Pu; 116, 280 and 579 g/t IHM of Am after 0, 3 and 10 years, respectively.

21.3.3. Decay heat and physical properties

As the radioactivity of the FP and actinides decreases by time, so does the energy absorbed in the shielding material (and by self-absorption) which is seen as *decay heat*. Table 21.5 gives data on the decay heat with contributions separately for the fission products and the actinides. Figure 21.9 shows the variation of the average energy per decay with time. The maxima and minima in the total and actinide curves in Figure 21.9 are caused by the presence of both α- and β,γ emitters with different activities and half-lives and also by the evolution in some decay chains. For cooling times $>10^3$ y the decay heat from the actinides and their daughters dominates.

The decay heat is considerable at short cooling times due to the very high decay rate (see Fig. 19.15). Before unloading spent fuel from a reactor, the used fuel elements are first allowed to cool in the reactor by forced circulation. Within a few weeks they are then transferred under water to the cooling basin at the reactor site for an additional cooling time, usually 6—12 months, after which they may be transferred to a central spent fuel storage facility. In the absence of such facilities, spent fuel elements can be stored in the reactor pools for many years. During this time the radiation level and heat production decrease considerably.

Table 21.5 Decay heat from unpartitioned fuel, fission products and heavy elements

	Decay heat (W/kg initial U)		
Cooling time	**Total**	**Fission products**	**Heavy elements**
1 d	54.3	38.7	15.6
90 d	16.2	13.0	3.27
180 d	11.6	9.20	2.42
1 y	7.67	5.93	1.73
5 y	2.98	1.70	1.28
10 y	2.17	1.12	1.05
100 y	0.28	0.016	0.27
10^3 y	0.13	0.000036	0.13
10^4 y	0.017	0.000034	0.017
10^5 y	0.0080	0.000020	0.0080
10^6 y	0.0021	0.0000007	0.0021

Limits: Fission products 28<Z<68, heavy element Z>79.
(Basic data as in Table 21.2)

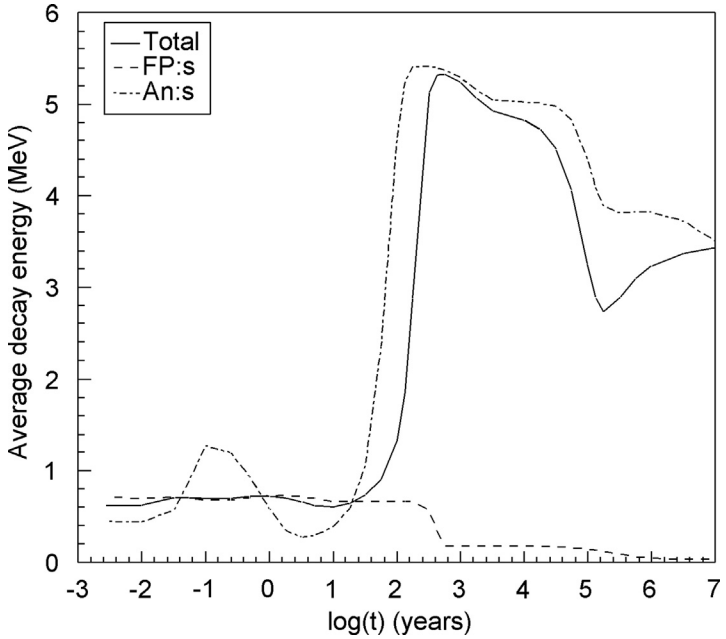


Figure 21.9 Average energy per decay in spent PWR fuel after 33 MWd/kg burnup.

21.4. MANAGEMENT OF SPENT FUEL

There are in principle several options for the handling of spent fuel. It can be:

a) Stored temporarily in pools, or dry storage, awaiting a decision on its future treatment

b) Reprocessed on order to recover fissionable material like U and Pu